

Catálogo de Átomos de Confusão em SystemVerilog

1.

Lambda - com smell

```
module mux (  
    input  logic [1:0] selector,  
    input  logic      enable,  
    output logic [7:0] out  
);  
  
    logic [7:0] decoded;  
  
    always_comb begin  
        case (selector)  
            0: decoded = 8'd10;  
            01: decoded = 8'd20;  
            10: decoded = 8'd30;  
            2: decoded = 8'd40;  
            default: decoded = 8'd100;  
        endcase  
  
        if (enable)  
            out = decoded;  
        else  
            out = 8'd0;  
        end  
    end  
endmodule  
  
selector = 10; enable = 1;
```

Gabarito:
out=40



Omega - sem smell

```
module mux (  
    input  logic [1:0] selector,  
    input  logic      enable,  
    output logic [7:0] out  
);  
  
    logic [7:0] decoded;  
  
    always_comb begin  
        case (selector)  
            2'b00: decoded = 8'd10;  
            2'b01: decoded = 8'd20;  
            2'b10: decoded = 8'd30;  
            2'b11: decoded = 8'd40;  
            default: decoded = 8'd100;  
        endcase  
  
        if (enable)  
            out = decoded;  
        else  
            out = 8'd0;  
        end  
    end  
endmodule  
  
selector = 10; enable = 1;
```

Gabarito:
out=30

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2.

Lambda - com smell

```
module top (  
    output logic [31:0] out0,  
    output logic [31:0] out1  
);  
  
    logic [1:0][31:0] A;  
  
    initial begin  
        A = '{1'b1, 1'b1};  
  
        out0 = A[0];  
        out1 = A[1];  
    end  
  
endmodule
```

Gabarito:

```
out0 = 00000000000000000000000000000001  
out1 = 00000000000000000000000000000001
```

Omega - sem smell

```
module top (  
    output logic [31:0] out0,  
    output logic [31:0] out1  
);  
  
    logic [1:0][31:0] A;  
  
    initial begin  
        A[0] = 1'b1;  
        A[1] = 1'b1;  
  
        out0 = A[0];  
        out1 = A[1];  
    end  
  
endmodule
```

Gabarito:

```
out0 = 00000000000000000000000000000001  
out1 = 00000000000000000000000000000001
```



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3.

Lambda - com smell

```
module example;

function int maxx(int a, int b);
    int max = a;

    if (b > max)
        max = b;

    return max;
endfunction

initial begin
    int r1, r2, r3;

    r1 = maxx(3, 7);
    r2 = maxx(1, 2);
    r3 = maxx(0, 0);
end

endmodule
```

Gabarito:

maxx(3, 7) = r1 = 7
maxx(1, 2) = r2 = 2
maxx(0, 0) = r3 = 0

Omega - sem smell

```
module example;

function automatic int maxx(int a, int b);
    int max = a;

    if (b > max)
        max = b;

    return max;
endfunction

initial begin
    int r1, r2, r3;

    r1 = maxx(3, 7);
    r2 = maxx(1, 2);
    r3 = maxx(0, 0);
end

endmodule
```

Gabarito:

maxx(3, 7) = r1 = 7
maxx(1, 2) = r2 = 2
maxx(0, 0) = r3 = 0



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4.

Lambda - com smell

```
module example (  
    input logic enable,  
    output int    out  
);  
    int result;  
    byte in = -5;  
  
    always_comb begin  
        result = in + 1'b1;  
  
        if (enable)  
            out = result;  
        else  
            out = in;  
    end  
endmodule
```

Gabarito:
enable=1 -> out=252

Omega - sem smell

```
module example (  
    input logic enable,  
    output int    out  
);  
    int result;  
    byte in = -5;  
  
    always_comb begin  
        result = in + 1;  
  
        if (enable)  
            out = result;  
        else  
            out = in;  
    end  
endmodule
```

Gabarito:
enable=1 -> out=-4



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5.

Lambda - com smell

```
module example;  
  
    int lo, med, hi;  
    bit result;  
    int selected;  
  
    initial begin  
        lo = 20;  
        med = 224;  
        hi = 164;  
  
        result = (lo < med < hi);  
  
        if (result)  
            selected = med;  
        else  
            selected = hi;  
    end  
endmodule
```

Gabarito:
result = 1, selected = 224

Omega - sem smell

```
module example;  
  
    int lo, med, hi;  
    bit result;  
    int selected;  
  
    initial begin  
        lo = 20;  
        med = 224;  
        hi = 164;  
  
        result = (lo < med) && (med < hi);  
  
        if (result)  
            selected = med;  
        else  
            selected = hi;  
    end  
endmodule
```

Gabarito:
result = 0, selected = 164



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6.

Lambda - com smell

```
module example (  
    input logic [3:0] instruction,  
    output logic [2:0] opcode  
);  
  
    always_comb begin  
        casex (instruction)  
  
            4'b0???:  
                opcode = 3'b001;  
  
            4'b1000:  
                opcode = 3'b010;  
  
            default:  
                opcode = 3'b111;  
  
        endcase  
    end  
endmodule
```

Gabarito:
instruction = 4'bxxxx
opcode = 3'b001

Omega - sem smell

```
module example (  
    input logic [3:0] instruction,  
    output logic [2:0] opcode  
);  
  
    always_comb begin  
        case (instruction) inside  
  
            4'b0???:  
                opcode = 3'b001;  
  
            4'b1000:  
                opcode = 3'b010;  
  
            default:  
                opcode = 3'b111;  
  
        endcase  
    end  
endmodule
```

Gabarito:
instruction = 4'bxxxx
opcode = 3'b111

